all-sites-exploratory

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Packages

```
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(scales)
library(readxl)
library(patchwork)
library(ggridges)
library(lme4)
library(lmerTest)
library(knitr)
library(rmarkdown)

library(ragg)
```

Data

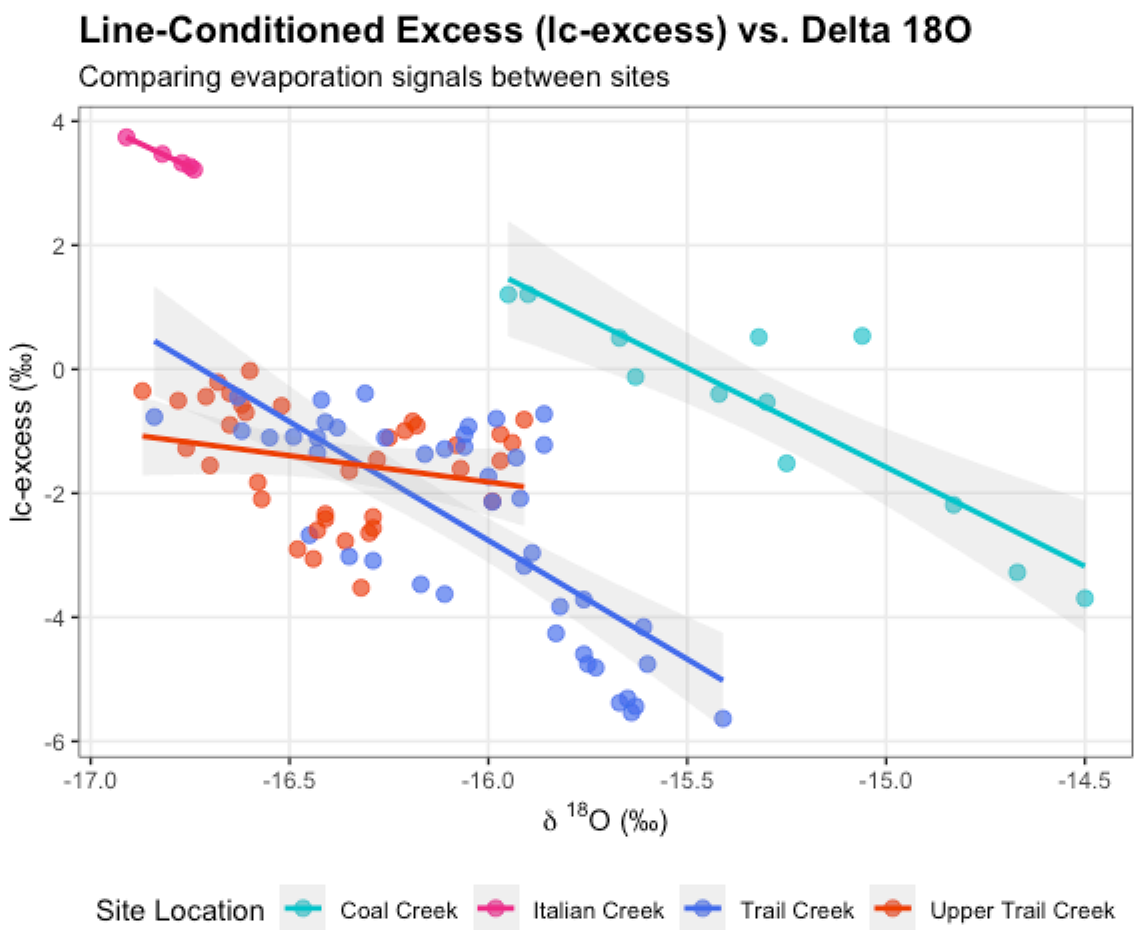
```
# Trail Creek Isotope
compiled_isotope                    <-                    read_csv(here("data",
"compiled_river_reach_isotope_data.csv"), show_col_types = FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stdev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stdev) |>
  mutate(date = mdy(date)) # telling r the date format


gradient_data <- read_csv(here("data", "tc_gradient_data.csv"))

trail_creek <- compiled_isotope |>
  filter(location %in% c("utc", "tc"))
```

```
coal_creek <- compiled_isotope |>
  filter(location == "cc")

italian_creek <- compiled_isotope |>
  filter(location == "ic")
```

Exploratory Visuals with ALL Sites

Line-Conditioned Excess (lc-excess) vs Delta 18O

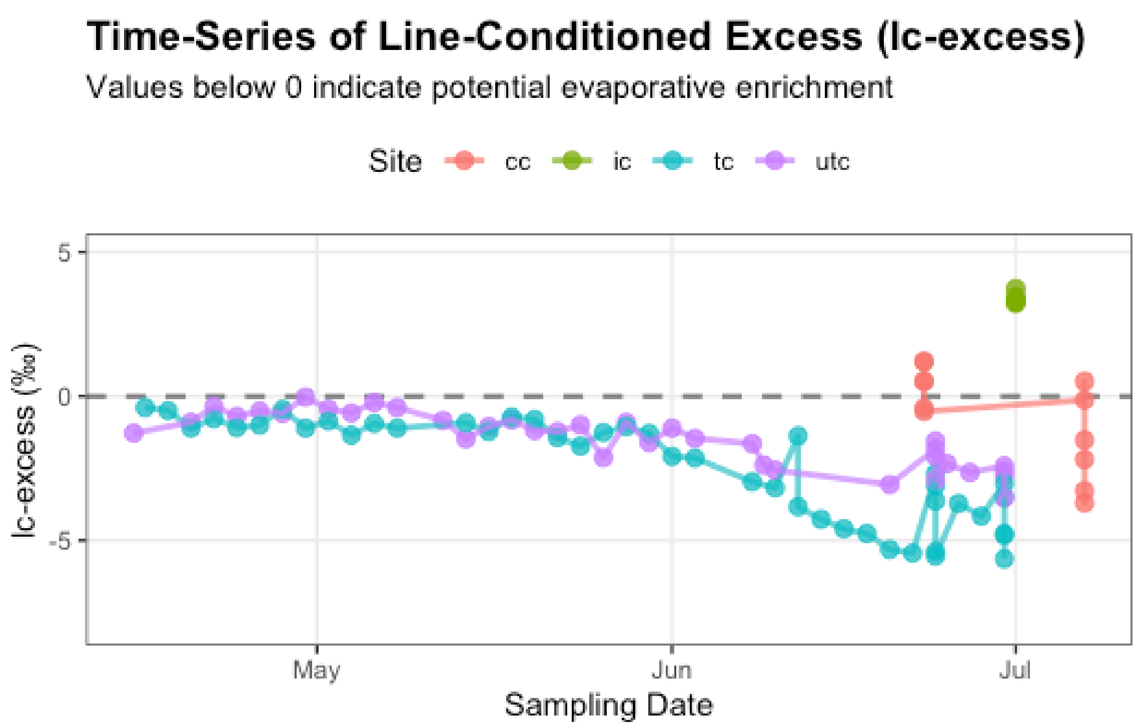
```
ggplot(data = compiled_isotope, aes(x = delta_o, y = lc_excess, color = location))
+
  geom_point(size = 2.5, alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15, linewidth = 1) +
  scale_color_manual(
     values = c("tc" = "royalblue2", "cc" = "turquoise3", "ic" = "deeppink2",
"utc" = "orangered2"),
    labels = c("cc" = "Coal Creek", "ic" = "Italian Creek", "tc" = "Trail Creek",
"utc" = "Upper Trail Creek")
  ) +
  labs(
    title = "Line-Conditioned Excess (lc-excess) vs. Delta 18O",
    subtitle = "Comparing evaporation signals between sites",
    x = expression(paste(delta, " "^18, "O (\u2030)")), # Formats as δ18O (‰)
    y = "lc-excess (\u2030)",
    color = "Site Location") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "bottom",
        panel.grid.minor = element_blank())
```



Time-Series of Line-Conditioned Excess (lc-excess)

```
ggplot(data = compiled_isotope,
       aes(x = date,
           y = lc_excess,
           color = location,
           group = location)) +
  geom_hline(yintercept = 0,
             linetype = "dashed",
             color = "gray50",
             size = 0.8) +
  geom_point(size = 2.5,
             alpha = 0.8) +
  geom_line(linewidth = 1,
            alpha = 0.7) +
  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
       subtitle = "Values below 0 indicate potential evaporative enrichment",
```

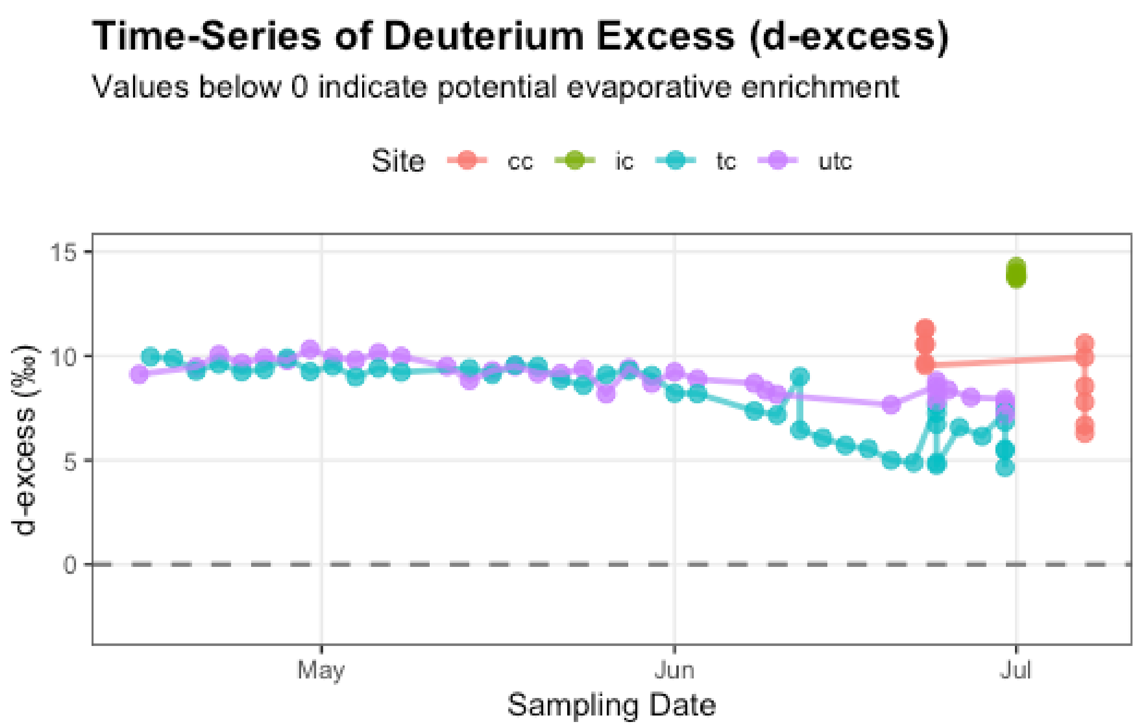
```
      x = "Sampling Date",
      y = "lc-excess (\u2030)",
      color = "Site") +
  ylim(-8,5) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        legend.position = "top")
```



Time-Series of Deuterium Excess (d-excess)

```
ggplot(data = compiled_isotope,
       aes(x = date,
           y = d_excess,
           color = location,
           group = location)) +
  geom_hline(yintercept = 0,
             linetype = "dashed",
             color = "gray50",
             size = 0.8) +
  geom_point(size = 2.5,
             alpha = 0.8) +
```

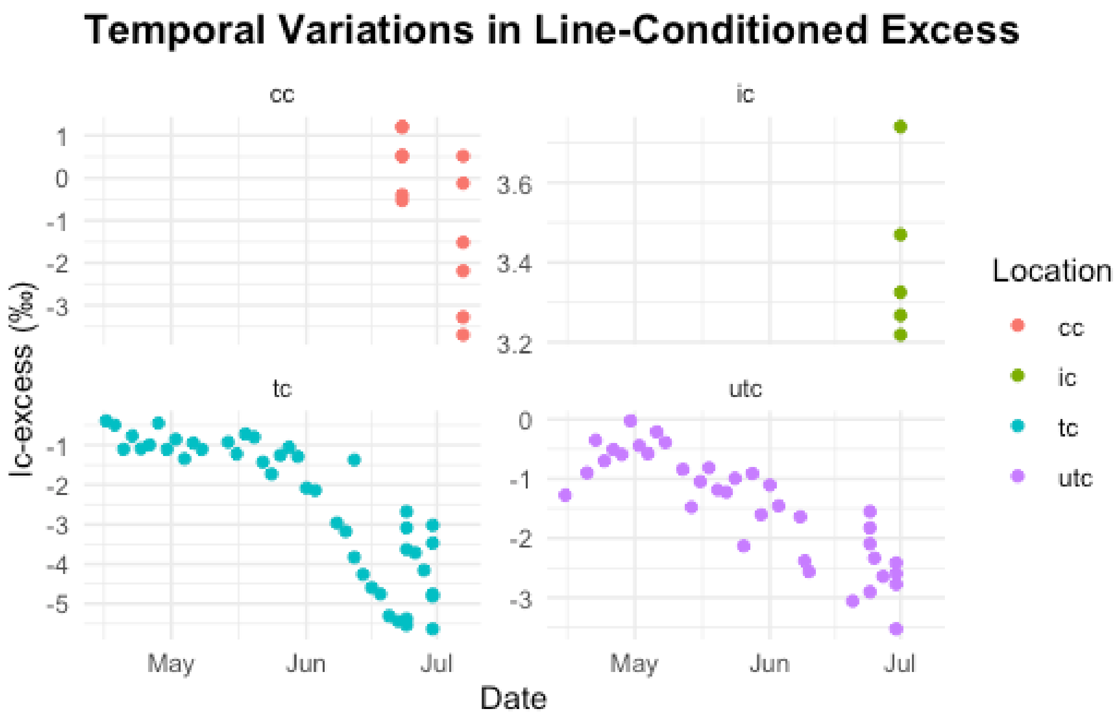
```
  geom_line(size = 1,
            alpha = 0.7) +
  labs(title = "Time-Series of Deuterium Excess (d-excess)",
       subtitle = "Values below 0 indicate potential evaporative enrichment",
       x = "Sampling Date",
       y = "d-excess (\u2030)",
       color = "Site") +
  ylim(-3,15) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        legend.position = "top")
```



Temporal Variations in Line-Conditioned Excess (by location)

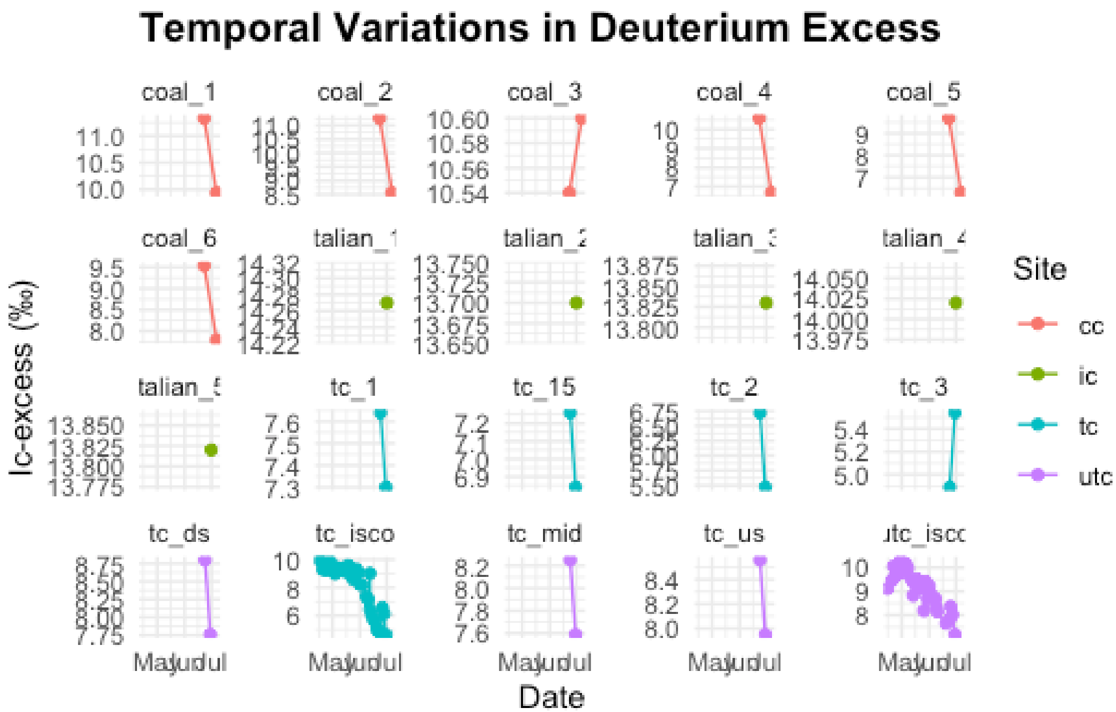
```
ggplot(compiled_isotope, aes(x = date, y = lc_excess, color = location)) +
  geom_point(size = 1.5) +
  labs(
    title = "Temporal Variations in Line-Conditioned Excess",
    x = "Date",
    y = "lc-excess (‰)",
    color = "Location"
```

```
  ) +
  facet_wrap(~location, scales = "free_y") +  # Compare across sites easily
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```



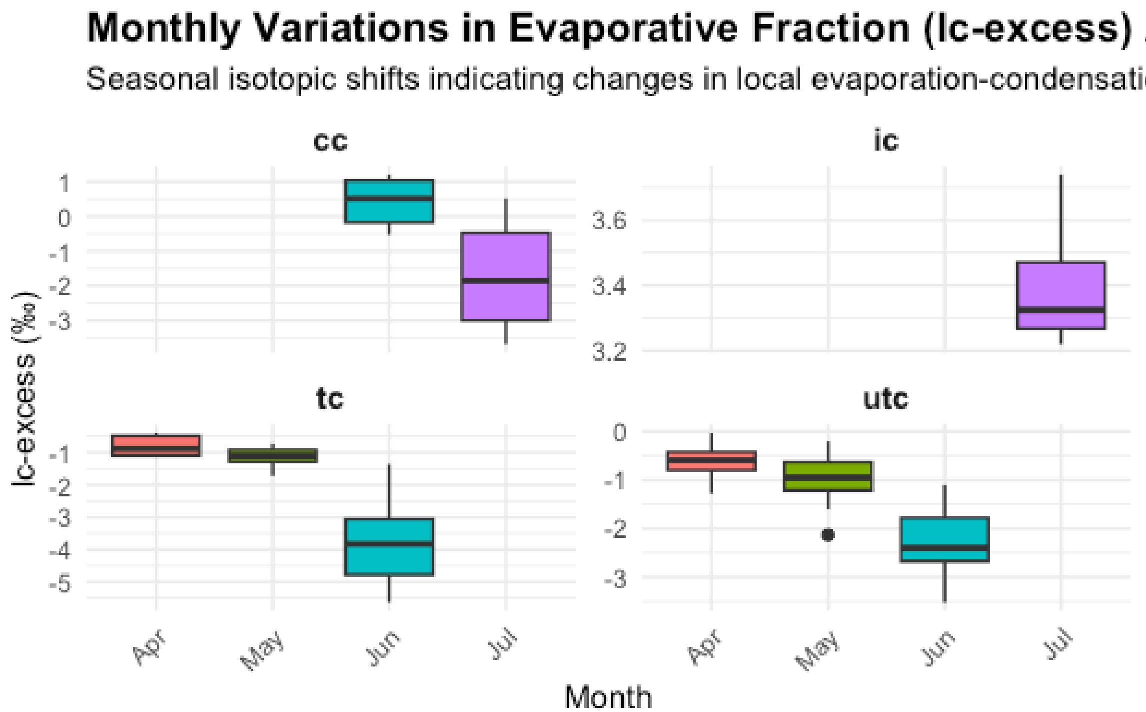
Temporal Variations in Deuterium Excess (by sites)

```
ggplot(compiled_isotope, aes(x = date, y = d_excess, color = location)) +
  geom_line(linewidth = 0.5) +
  geom_point(size = 1.5) +
  labs(
    title = "Temporal Variations in Deuterium Excess",
    x = "Date",
    y = "lc-excess (‰)",
    color = "Site"
  ) +
  facet_wrap(~site, scales = "free_y") +  # Compare across sites easily
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```



Monthly Variations in Evaporative Fraction (lc-excess) Across Study Sites

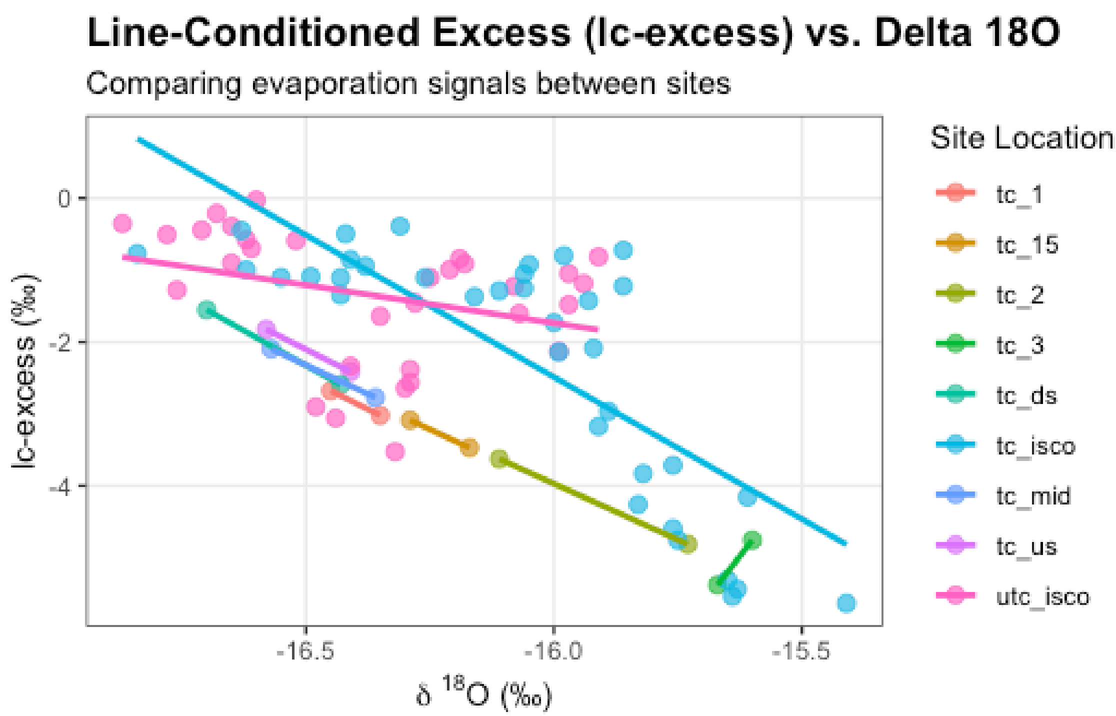
```
ggplot(compiled_isotope, aes(x = factor(month(date, label = TRUE, abbr = TRUE)),
y = lc_excess)) +
  geom_boxplot(aes(fill = factor(month(date))), show.legend = FALSE) +
  facet_wrap(~location, scales = "free_y") +
 labs(
     title = "Monthly Variations in Evaporative Fraction (lc-excess) Across
Study Sites",
    subtitle = "Seasonal isotopic shifts indicating changes in local evaporation-
condensation histories",
    x = "Month",
    y = "lc-excess (‰)"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14),
    strip.text = element_text(face = "bold", size = 11),
    axis.text.x = element_text(angle = 45, hjust = 1)) # tilting the x axis text
```



Exploratory Visuals of Trail Creek

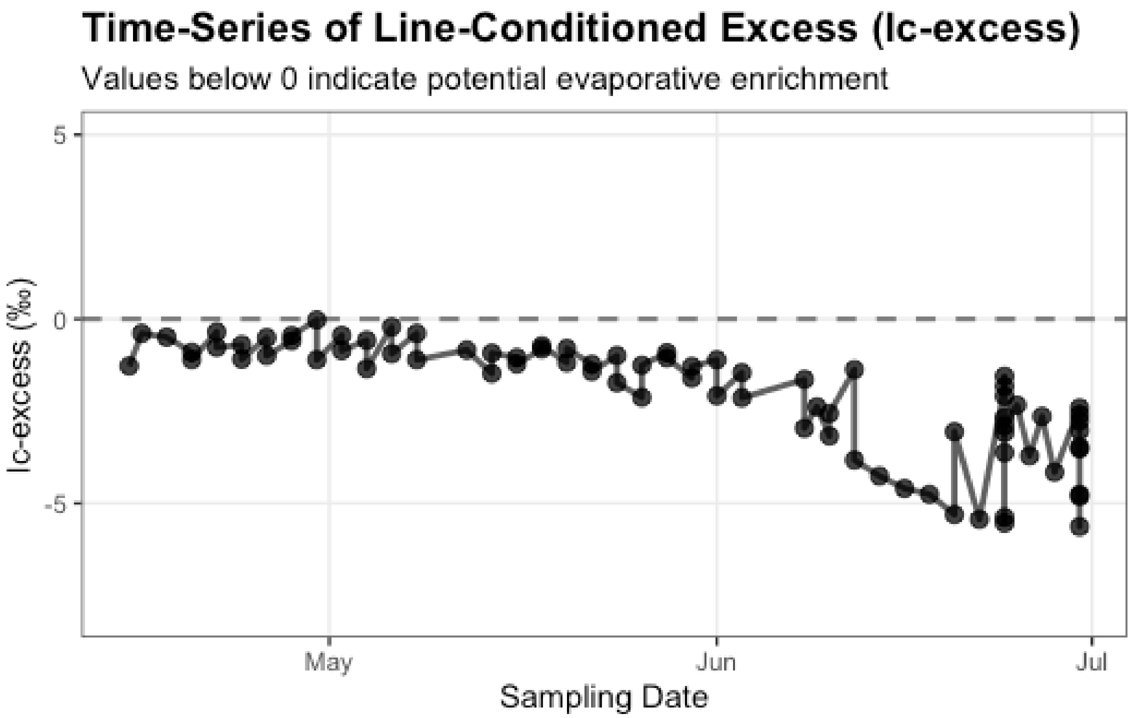
Line-Conditioned Excess (lc-excess) vs. Delta 18O

```r
ggplot(data = trail_creek, aes(x = delta_o, y = lc_excess, color = site)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE, alpha = 0.15, linewidth = 1) +
  labs(
    title = "Line-Conditioned Excess (lc-excess) vs. Delta 18O",
    subtitle = "Comparing evaporation signals between sites",
    x = expression(paste(delta, " "^18, "O (\u2030)")), # Formats as δ18O (‰)
    y = "lc-excess (\u2030)",
    color = "Site Location") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "right",
        panel.grid.minor = element_blank())
```



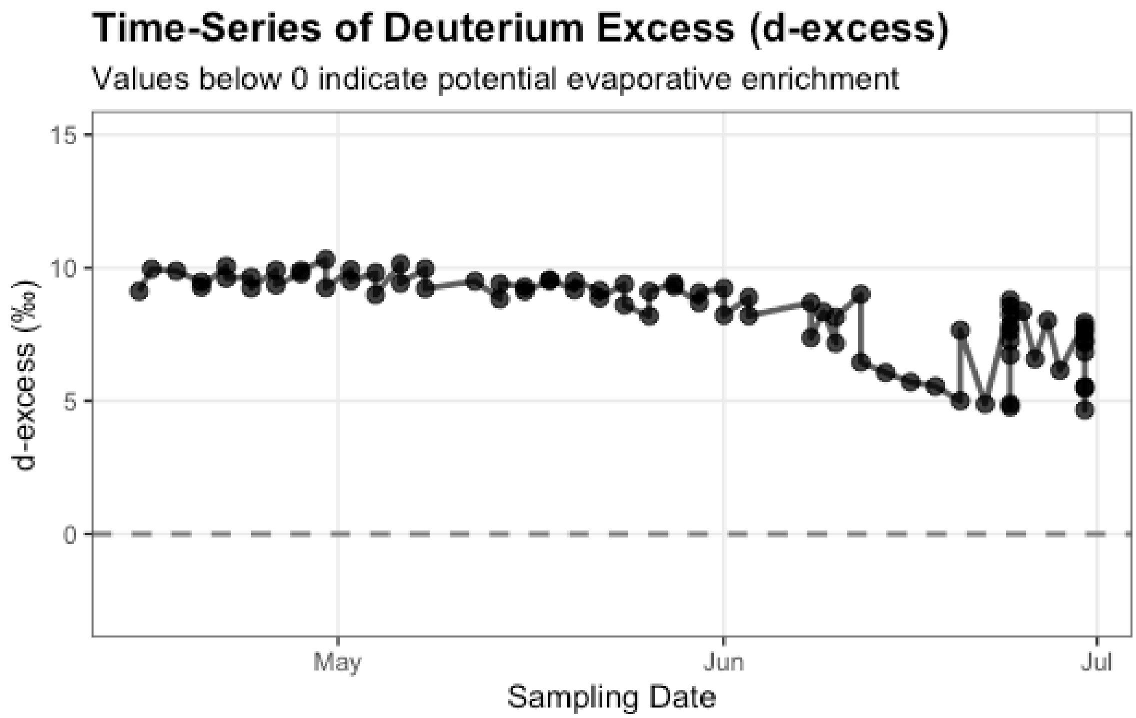
Time-Series of Line-Conditioned Excess (lc-excess)

```
ggplot(data = trail_creek,
       aes(x = date,
           y = lc_excess)) +
  geom_hline(yintercept = 0,
             linetype = "dashed",
             color = "gray50",
             size = 0.8) +
  geom_point(size = 2.5,
             alpha = 0.8) +
  geom_line(size = 1,
            alpha = 0.7) +
  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
       subtitle = "Values below 0 indicate potential evaporative enrichment",
       x = "Sampling Date",
       y = "lc-excess (\u2030)",
       color = "Site") +
  ylim(-8,5) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        legend.position = "top")
```



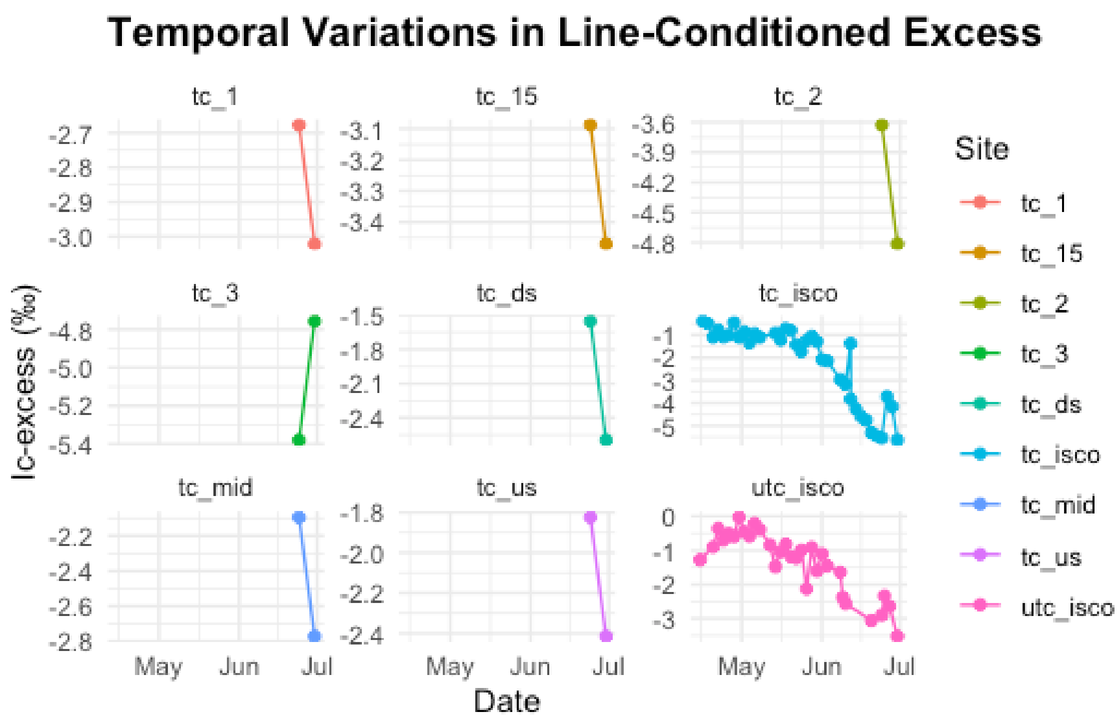
Time-Series of Deuterium Excess (d-excess)

```
ggplot(data = trail_creek,
       aes(x = date,
           y = d_excess)) +
  geom_hline(yintercept = 0,
             linetype = "dashed",
             color = "gray50",
             size = 0.8) +
  geom_point(size = 2.5,
             alpha = 0.8) +
  geom_line(size = 1,
            alpha = 0.7) +
  labs(title = "Time-Series of Deuterium Excess (d-excess)",
       subtitle = "Values below 0 indicate potential evaporative enrichment",
       x = "Sampling Date",
       y = "d-excess (\u2030)",
       color = "Site") +
  ylim(-3,15) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        legend.position = "top")
```



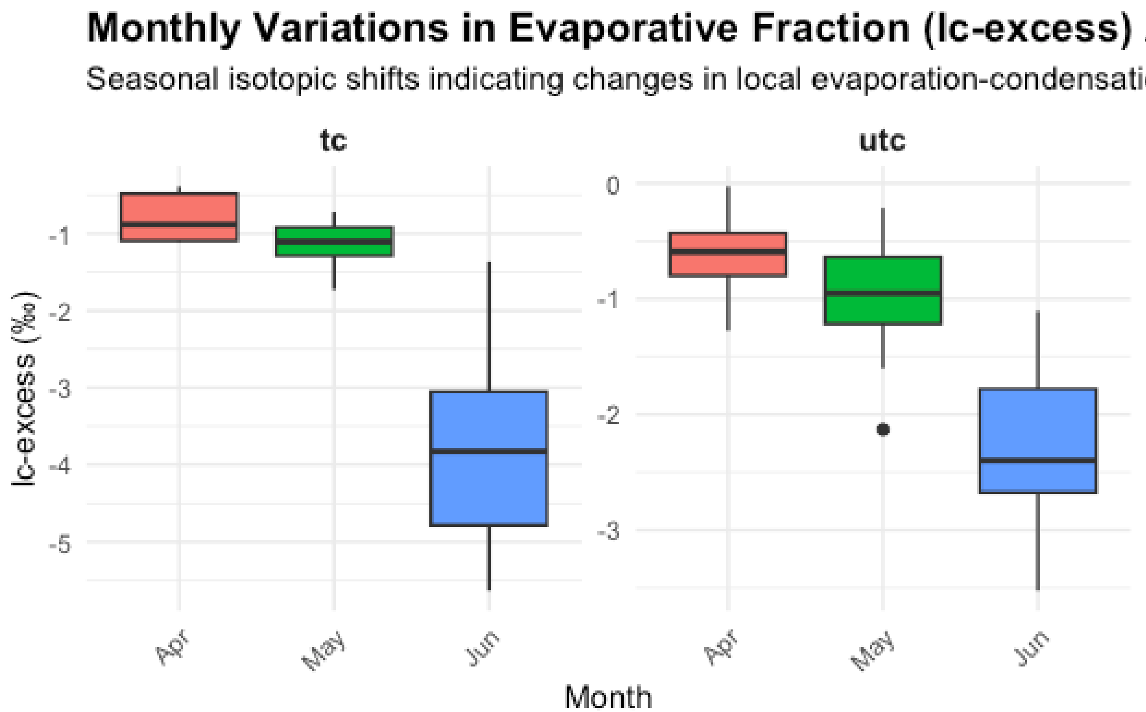
Temporal Variations in Line-Conditioned Excess

```
ggplot(trail_creek, aes(x = date, y = lc_excess, color = site)) +
  geom_line(linewidth = 0.5) +
  geom_point(size = 1.5) +
  labs(
    title = "Temporal Variations in Line-Conditioned Excess",
    x = "Date",
    y = "lc-excess (‰)",
    color = "Site"
  ) +
  facet_wrap(~site, scales = "free_y") +  # Compare across sites easily
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```



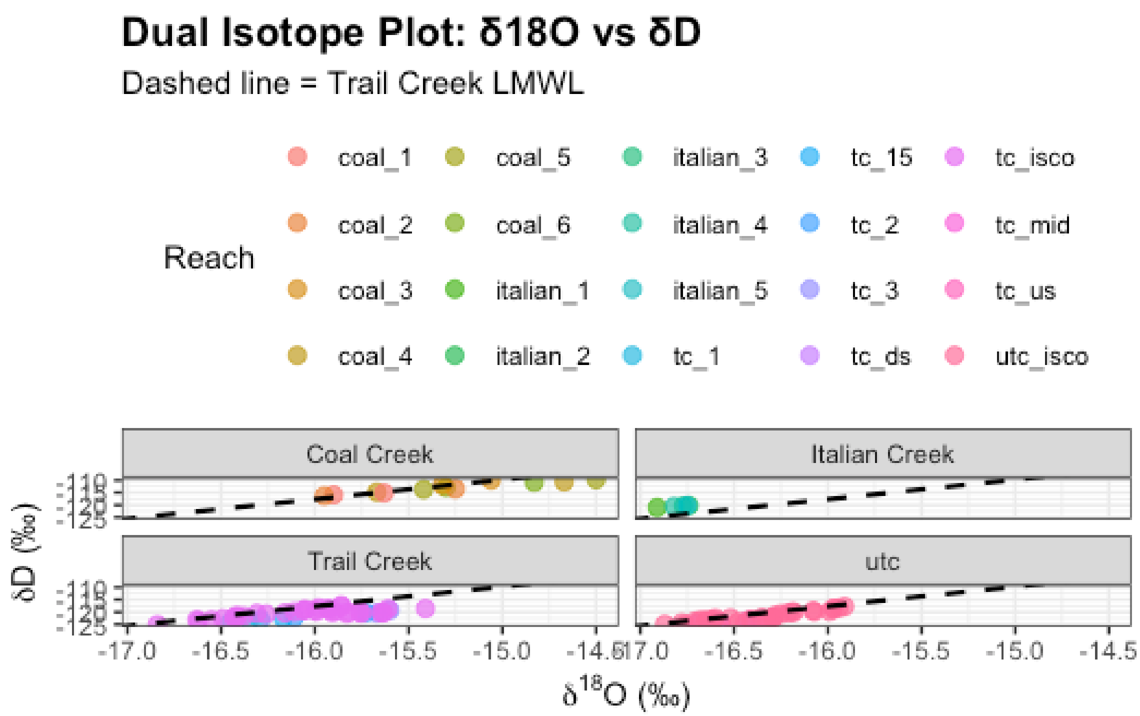
Monthly Variations in Evaporative Fraction (lc-excess) Across Study Sites

```
ggplot(trail_creek, aes(x = factor(month(date, label = TRUE, abbr = TRUE)), y =
lc_excess)) +
  geom_boxplot(aes(fill = factor(month(date))), show.legend = FALSE) +
  facet_wrap(~location, scales = "free_y") +
 labs(
     title = "Monthly Variations in Evaporative Fraction (lc-excess) Across
Study Sites",
    subtitle = "Seasonal isotopic shifts indicating changes in local evaporation-
condensation histories",
    x = "Month",
    y = "lc-excess (‰)"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14),
    strip.text = element_text(face = "bold", size = 11), # Emphasize site names
    axis.text.x = element_text(angle = 45, hjust = 1)   # Tilt month labels if
they overlap
  )
```



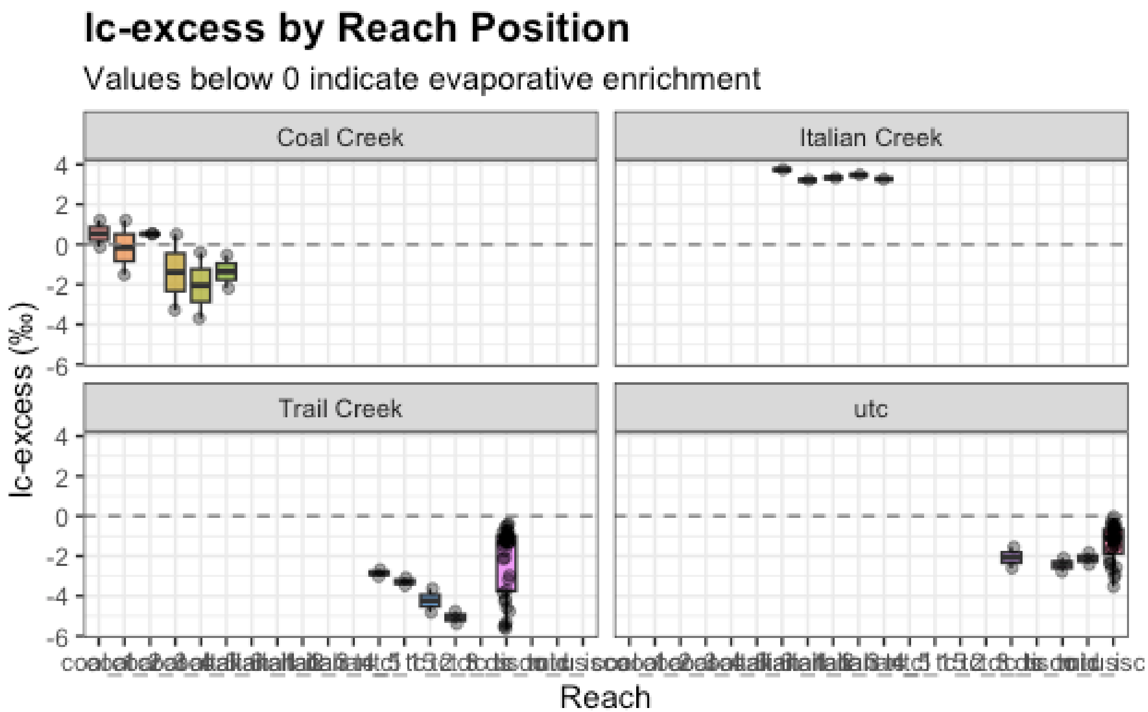
Dual-isotope plot (δ18O vs δD) with LMWL reference

```
# Trail Creek LMWL calculated from OPIC

lmwl_slope <- 7.94
lmwl_intercept <- 9.37

ggplot(compiled_isotope, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_abline(slope = lmwl_slope, intercept = lmwl_intercept,
              linetype = "dashed", color = "black", linewidth = 0.8) +
  facet_wrap(~location, labeller = labeller(location = c(tc = "Trail Creek",
                                                         cc = "Coal Creek",
                                                            ic = "Italian
Creek"))) +
  labs(title = "Dual Isotope Plot: \u03b418O vs \u03b4D",
       subtitle = "Dashed line = Trail Creek LMWL",
       x = expression(paste(delta^18, "O (\u2030)")),
       y = expression(paste(delta, "D (\u2030)")),
       color = "Reach") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "top")
```



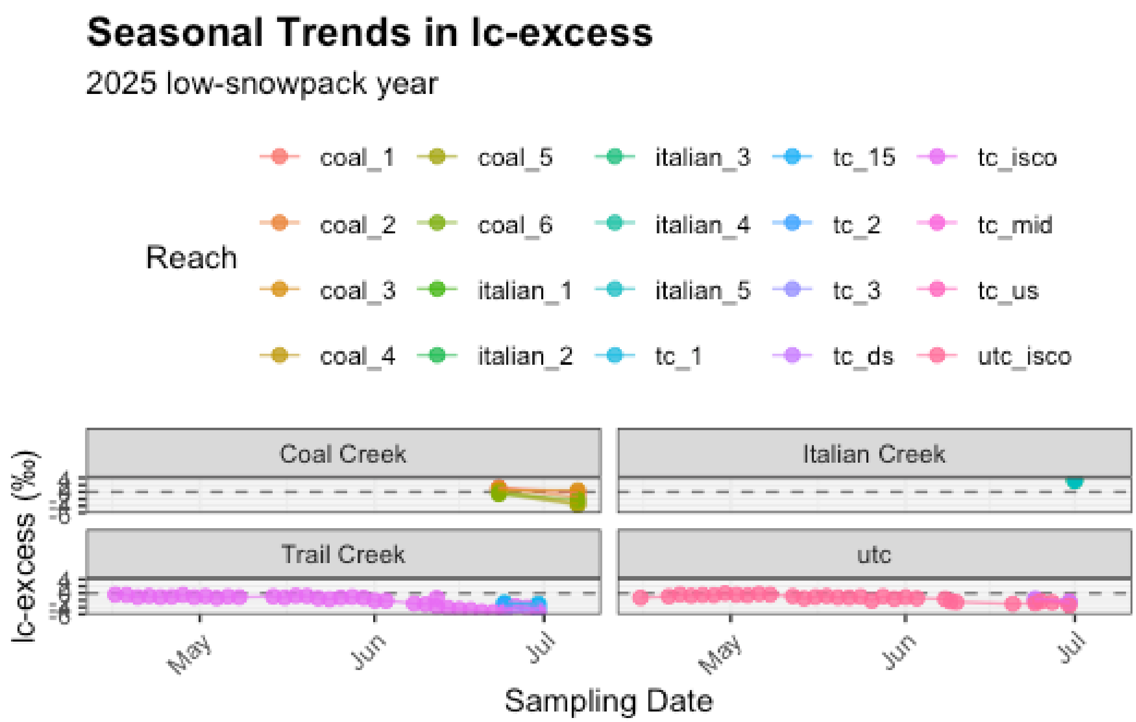
Lc-excess boxplot by reach, faceted by watershed

```
ggplot(compiled_isotope, aes(x = site, y = lc_excess, fill = site)) +
  geom_boxplot(alpha = 0.7, outlier.shape = 21) +
  geom_jitter(width = 0.15, alpha = 0.4, size = 1.5) +
  facet_wrap(~location, labeller = labeller(location = c(tc = "Trail Creek",
                                                           cc = "Coal Creek",
                                                                 ic = "Italian
Creek"))) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  labs(title = "lc-excess by Reach Position",
       subtitle = "Values below 0 indicate evaporative enrichment",
       x = "Reach", y = "lc-excess (\u2030)") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "none")
```



Time series of lc-excess across season, colored by reach, faceted by watershed

```
ggplot(compiled_isotope, aes(x = date, y = lc_excess, color = site, group =
interaction(site, location))) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  geom_point(size = 2, alpha = 0.8) +
  geom_line(alpha = 0.6) +
  facet_wrap(~location, labeller = labeller(location = c(tc = "Trail Creek",
                                                          cc = "Coal Creek",
                                                               ic = "Italian
Creek"))) +
  labs(title = "Seasonal Trends in lc-excess",
       subtitle = "2025 low-snowpack year",
       x = "Sampling Date", y = "lc-excess (\u2030)", color = "Reach") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "top",
        axis.text.x = element_text(angle = 45, hjust = 1))
```



Mixed model: lc-excess ~ site * location, with sample as random effect if repeated

```
# Mixed model
model_mixed <- lmer(lc_excess ~ site * location + (1 | date), data =
compiled_isotope)
summary(model_mixed)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: lc_excess ~ site * location + (1 | date)
   Data: compiled_isotope

REML criterion at convergence: 257.6

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.97083 -0.32109  0.04244  0.42476  1.72105

Random effects:
 Groups   Name        Variance Std.Dev.
 date     (Intercept) 1.6758   1.2945
 Residual             0.4748   0.6891
```

```
Number of obs: 98, groups:  date, 43

Fixed effects:
              Estimate Std. Error       df t value Pr(>|t|)    
(Intercept)    0.54145    1.03697 45.43586   0.522 0.604105    
sitecoal_2    -0.69845    0.68906 36.35111  -1.014 0.317458    
sitecoal_3    -0.01475    0.68906 36.35111  -0.021 0.983039    
sitecoal_4    -1.92835    0.68906 36.35111  -2.799 0.008163 ** 
sitecoal_5    -2.58865    0.68906 36.35111  -3.757 0.000603 ***
sitecoal_6    -1.90100    0.68906 36.35111  -2.759 0.009026 ** 
siteitalian_1  3.19845    1.79609 45.43586   1.781 0.081633 .  
siteitalian_2  2.67715    1.79609 45.43586   1.491 0.142991    
siteitalian_3  2.78385    1.79609 45.43586   1.550 0.128091    
siteitalian_4  2.92835    1.79609 45.43586   1.630 0.109930    
siteitalian_5  2.72605    1.79609 45.43586   1.518 0.136000    
sitetc_1      -1.22304    1.20615 65.40299  -1.014 0.314317    
sitetc_15     -1.65284    1.20615 65.40299  -1.370 0.175265    
sitetc_2      -2.59424    1.20615 65.40299  -2.151 0.035189 *  
sitetc_3      -3.44214    1.20615 65.40299  -2.854 0.005779 ** 
sitetc_ds     -0.44794    1.20615 65.40299  -0.371 0.711554    
sitetc_isco   -2.78707    1.06395 45.62132  -2.620 0.011914 *  
sitetc_mid    -0.80694    1.20615 65.40299  -0.669 0.505836    
sitetc_us     -0.49374    1.20615 65.40299  -0.409 0.683618    
siteutc_isco  -2.14744    1.06549 45.84336  -2.015 0.049742 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
fit warnings:
fixed-effect model matrix is rank deficient so dropping 60 columns / coefficients
```

```r
anova(model_mixed)  # Type III tests via lmerTest
```

```
Type III Analysis of Variance Table with Satterthwaite's method
              Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)    
site          44.331  2.3332    19 38.53  4.9141 1.402e-05 ***
location                                                      
site:location                                                 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

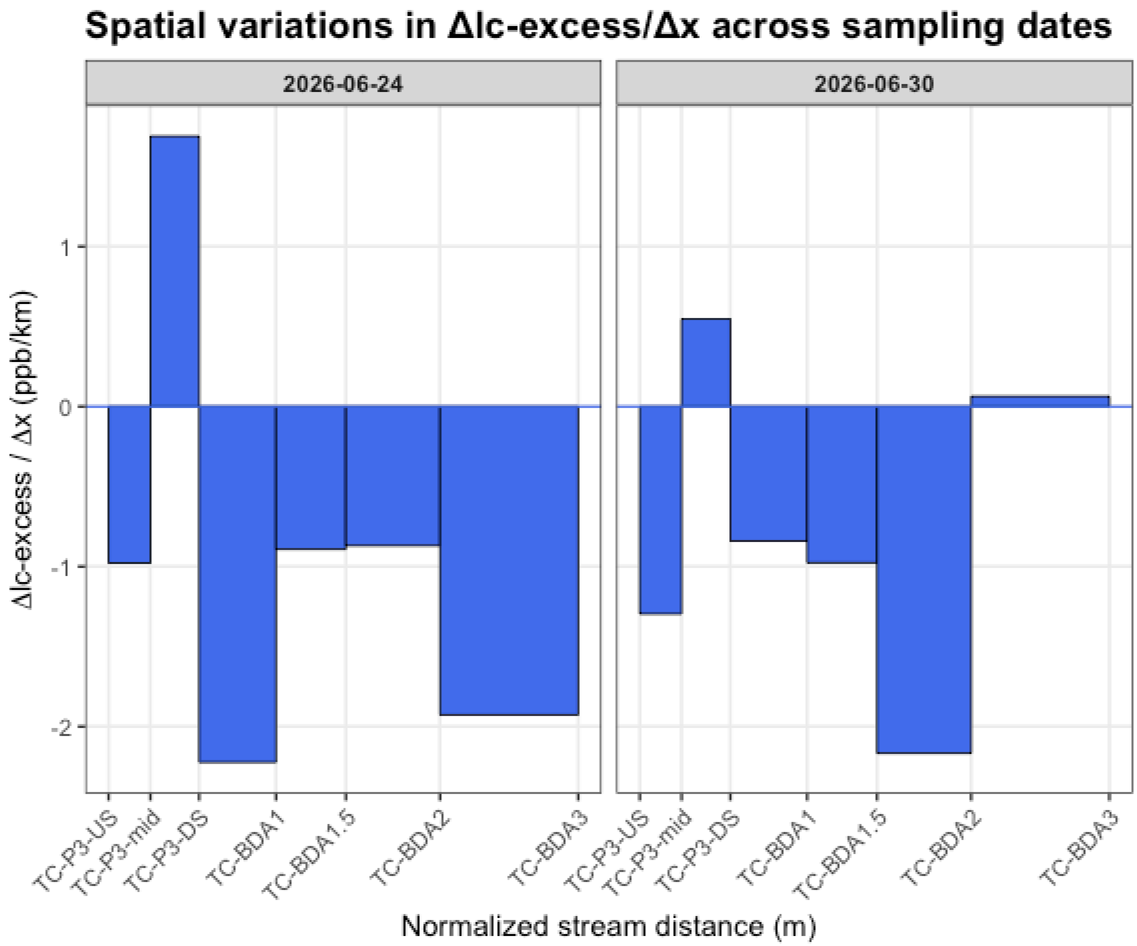
```r
ggplot(compiled_isotope, aes(x = location, y = lc_excess, color = site, group =
site)) +
  stat_summary(fun = mean, geom = "point", size = 3) +
  stat_summary(fun = mean, geom = "line", linewidth = 1) +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.1) +
  labs(title = "Interaction Plot: Reach \u00d7 Watershed",
       x = "Watershed", y = "Mean lc-excess (\u2030)", color = "Reach") +
  theme_bw()
```


```r
delta_summary <- compiled_isotope |>
  group_by(site) |>
  summarize(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )


ggplot(delta_summary, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_errorbar(
    aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
```

```
    width = 0.3,
    color = "black"
  ) +
  theme_minimal() +
  labs(
    title = "Mean LC Excess by Site with Standard Error",
    x = "Sampling Site",
    y = "Mean LC Excess"
  ) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1), # Tilts labels so they
don't overlap
    legend.position = "none"
  )
```


```
# Buildings a look up table of stream distances
distances <- tibble(site = c("TC-P3-US", "TC-P3-mid", "TC-P3-DS", "TC-BDA1",
                             "TC-BDA1.5", "TC-BDA2", "TC-BDA3"),
                    dist_m = c(57.6, 333, 654, 1160, 1620, 2240, 3150)) |>
  mutate(xmax = dist_m,
         xmin = lag(dist_m, default = 0)) # previous site's distance, 0 for first
```

```r
# Creating a filtered data frame for lc-excess
gradient_lc_clean <- gradient_data |>
  select(site, normalized_stream_distance_m, `Δlc-excess/Δx_6-24`, `Δlc-excess/
Δx_6-30`) |>
  pivot_longer( cols = c(`Δlc-excess/Δx_6-24`, `Δlc-excess/Δx_6-30`),
                names_to = "date",
                values_to = "gradient") |>
  mutate(date = case_when(date == "Δlc-excess/Δx_6-24" ~ "2026-06-24",
                          date == "Δlc-excess/Δx_6-30" ~ "2026-06-30"),
         gradient = as.numeric(gradient)) |>
  left_join(distances %>% select(site, xmin, xmax), by = "site") |>
  mutate(site = factor(site, levels = distances$site)) |>
  filter(!is.na(gradient))

# Gradient plot
ggplot(gradient_lc_clean) +
  geom_rect(aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
            fill = "royalblue2", color = "black", linewidth = 0.3) +
  geom_hline(yintercept = 0, color = "royalblue2", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
  scale_x_continuous(
    breaks = distances$dist_m,
    labels = distances$site) +
  labs(title = "Spatial variations in Δlc-excess/Δx across sampling dates",
       x = "Normalized stream distance (m)",
       y = expression(paste(Delta, "lc-excess / ", Delta, "x (ppb/km)"))) +
  theme_bw(base_size = 11) +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        strip.text = element_text(face = "bold"),
        axis.text.x = element_text(angle = 45, hjust = 1))
```



All reaches show lc-excess decreasing downstream, on both dates. Since lc-excess becoming more negative = more evaporative enrichment, this version says every single reach along Trail Creek is adding evaporative signal, with no recovery zones. That's a materially different story than "some reaches enrich, some reaches dilute" — it now reads as continuous, cumulative evaporative loss along the whole restored gradient.

Where the deepest enrichment is happening differs by date:

6-24: TC-P3-DS reach is the sharpest dip (~−2.2), essentially a local hotspot just downstream of the mid-P3 restoration point. 6-30: TC-BDA1.5⟶BDA2 is now the sharpest dip (~−2.15), while TC-P3-DS has relaxed to a much smaller magnitude (~−0.55) compared to 6-24.

TC-BDA3 is the outlier reach on both dates — it's close to flat (near-zero gradient), meaning that final reach adds almost no additional evaporative signal, regardless of date. That's consistent between both panels and is probably your most robust finding here: whatever's happening upstream, the lowest reach isn't contributing much additional fractionation.

Exploratory Visuals of Coal Creek

Exploratory Visuals of Italian Creek